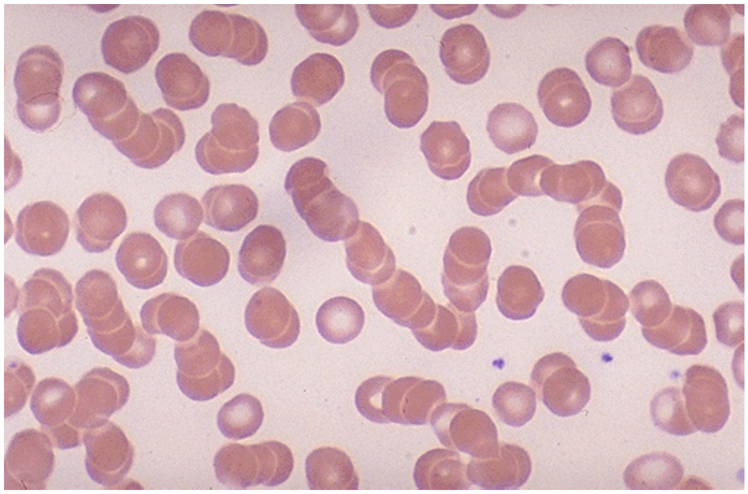
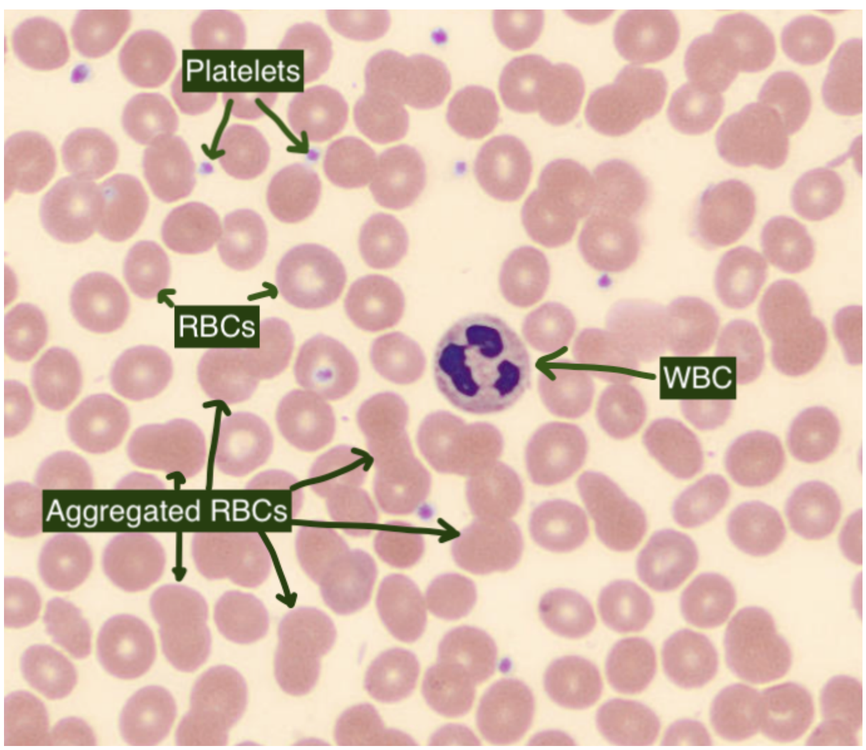
One condition we would like to detect on peripheral blood smear images is Rouleaux Formation in Red Blood Cells. Rouleaux formation refers to the stacking of red blood cells (RBCs) in a coin-like pattern, like a stack of rouleaux (French for "little wheels"):



Source: https://webpath.med.utah.edu/HEMEHTML/HEME007.html

To get to that goal, we initially want an algorithm to identify RBCs on blood smears, and then distinguish aggregated RBC clusters from single RBCs



Key expected outcomes with quantified evaluations

a. Identification of single RBCs
b. Identification of aggregated RBCs

    c. Segmenting individual cells and/or counting the number of cells in aggregated RBC clusters (or explain whether that's possible given the image samples provided and if not, what we need to make that possible)

To achieve this goal:

1. With the sample images included, you should create your own ground truth labels on some smaller crops of the images and use those to evaluate your algorithms.
2. Come up with appropriate ways to display, present the algorithm outputs (bounding boxes around objects, binary masks, overlaid masks on top of images, etc).
3. Choose appropriate evaluation metrics to gauge the performance of the object detection algorithm you develop.
4. You can identify previous work done in the field, get ideas from, and even use out-of-the-box solutions if available:
    a. https://lup.lub.lu.se/luur/download?func=downloadFile&recordOId=9066480&fileOId=9066481
    b. …
5. If you find related work in the literature that you use, justify the use and mention why a better solution could not be developed in the given time.
6. Develop your solution in Python, with your choice of related libraries (opencv, scipy, etc). The task is best suited to be tackled with classic computer vision methods such as thresholding, morphological operations like dilation, erosion, opening, and closing, edge detection, etc.
7. Try to incorporate software engineering best practices including object-oriented programming and ensure your code is well-documented and easy to set up and use by another person.